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SEAT No. :

P3660

[Total No. of Pages : 2

APR - 15 / Engg. - 145

T.E. (Computer Engineering) (In Sem - Semester - II)

DIGITAL SIGNAL PROCESSING APPLICATIONS

(2012 Pattern)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) *Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Assume suitable data, if necessary.*

Q1) a) Why the problem of aliasing is observed during the sampling process? Derive the relationship between analog frequency F and DT frequency f . **[5]**

b) State the advantages of Digital Signal Processing and give the classification of DT signals. **[5]**

OR

Q2) a) Define the impulse response of a system and obtain it for a system described as **[5]**

$$y(n) = x(n) + 0.5 y(n-1)$$

b) Represent a DT sequence $x(n)$ using convolution. Obtain the computational complexity of linear convolution of two DT signals having N samples each. **[5]**

Q3) a) State and prove : **[5]**

- i) time reversal and
- ii) time shifting property of a FT.

b) State the necessary condition for the existence of FT. Explain how the magnitude response of a FT is obtained? Obtain DT frequencies $\{\omega_k\}$ for 8-point DFT. **[5]**

OR

P.T.O.

- Q4)** a) Obtain 4 point DFT for a sequence $x(n) = \{1, -2, 2, 1\}$ and plot the magnitude spectrum. [5]
- b) Discuss the periodicity and circular shifting property of N point DFT. How convolution property of DFT is different than FT? [5]

- Q5)** a) Obtain ZT of a DT signal using ZT properties where, [5]
- $$x(n) = 2^n \cdot u(n-1)$$
- b) Derive the relationship between ZT and FT. [5]

OR

- Q6)** a) Obtain ZT of two standard signals $u(n)$ and $\delta(n)$. Plot its ROCs. [5]
- b) Define a system function $H(Z)$. How it describes the properties of DT system? What is pole zero plot? [5]

